

Eunyul (Estelle) Kim estellekimdev.com | (714) 719-5688 | kestelle@sas.upenn.edu

University of Pennsylvania, c/o 2027 GPA: 3.5 / 4.0 **Dual MSE + BSE:** CS + Computer Graphics + 3D Art

EXPERIENTIAL INTERFACES

Internet Atlas — Spatial Browsing Interface (Three.js, D3.js, React, FastAPI, Supabase, Pinecone) **2025**

Led development of a 3D spatial interface translating browsing behavior into an explorable force-directed environment.

- Built an embedding pipeline to encode visual + textual website signals into “ambiance” clustering logic.
- Engineered custom camera choreography, interaction states, and hybrid SVG/3D animation for fluid navigation.

Into the Blue — Interactive Museum Experience (React, Next.js) **2024**

- Built a camera-to-sticker pipeline blending live visitor input with generative visual layering.
- Architected offline-first, modular browser storage to accommodate spotty Wi-Fi and evolving exhibitions.
- Maintained a long-running deployment (9+ months) used by 180k+ visitors with stable performance.

Mini Minecraft — Real-Time Voxel Engine (C++, OpenGL, GLSL) **2024**

- Engineered a from-scratch voxel renderer with a custom OpenGL pipeline (team of 3).
- Implemented VBO-based terrain meshing, procedural sky shaders, and real-time water motion + reflections.

INTERACTIVE PROTOTYPES

Watercolor Shader — Real-Time Material Study (GLSL)

Ping-pong shader pipeline exploring bleed/drip behavior with interactive input and GPU-accelerated blending.

Softbody Jelly — Tactile Physics Prototype

3D soft-body deformation system optimized with 2D particle dynamics, procedural lathe meshing, and custom shading.

textellation — Generative Typographic Layout System

Organic word compositions using custom packing + force sim, treating text as a spatial system rather than static layout.

Magnetic Poetry — Real-Time Collaborative Interface

Multiuser drag-drop board with tactile interaction rules and WebSocket sync to support shared, live arrangement.

WORK EXPERIENCE & ACTIVITIES

Data Visualization & Engineering Intern @ Southern California Edison — Rosemead, CA **2025**

- Deployed and optimized ETL pipelines serving a utility with 15M residents, positive feedback & return offer
- Worked directly with cross-team stakeholders to deliver 10+ production data solutions under tight timelines.

Frontend Web Developer @ Children’s Hospital of Philadelphia — Philadelphia, PA **2025~**

- Developing ML-driven research management hub for autism studies using React, TypeScript, and REST APIs.
- Implemented scalable data visualization & state management components to streamline researcher workflow.

Vice President (External), 3x Project Lead, Full-Stack Developer @ UPenn Spark **2024~**

- Manage corporate relationships and partnerships for Penn’s largest community of creative devs and designers.
- Led redesign and rebuild of Spark’s website, improving information architecture, branding, and maintainability.
- Led 3 full-stack product teams; scoped features, shipped apps to hundreds / thousands of users.

Governmental Affairs Director @ California Association of Student Councils **2024**

- Authored & passed 3 CA laws, facilitated a 20-organization coalition including ACLU and Human Rights Watch.

SKILLS

Languages: Python, C++, Java, TypeScript, JavaScript, SQL, GLSL, HTML/CSS

Frameworks: React, OpenGL, WebGL, Next.js, React Native, Node.js, Django, PostgreSQL, MySQL, Supabase

Concepts: Rendering Pipelines, GPU Programming, Computer Graphics, Real-Time Systems, Data Pipelines, Machine Learning, Full-Stack Development, Systems Programming, Design Engineering, Prototyping, Interactive Interfaces

Tools: Figma, Adobe Creative Suite, Maya, Docker, AWS, Github, Git, Arduino (Electronics), Blender, TouchDesigner